

Bonus Project: System Parameter Identification with GenAI Assistants

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The lab report is due on **5/6, 11:59pm**. The experimental part is scripted in an .ipynb file attached. This project will be counted as raw 5pts bonus towards your final score. To complete this project, you need to install Python¹ and Jupyter Notebook² and know how to run an .ipynb file in Windows or Mac OS; ask any AI tools (e.g, ChatGPT) for instructions.

Background and objectives

Signals that are not grounded in physical systems are not reliable. Identifying the underlying system from observed signals is a fundamental problem in signal processing. In practice, exploring all possible system models requires significant time and computational resources. In this lab, you will explore how to identify the system models from observed input and output signals using the standard methods with an assistance of GenAI tools.

The underlying physical system is a series RC circuit (see below) with a resistor and a capacitor, where R is the resistance and C is the capacitance.

Perhaps you are already familiar with in class, we cast a series RC circuit as a system by selecting an input $x(t)$ and an output $y(t)$ below,

- Input signal $x(t) = v(t)$;
- Output signal $y(t) = v_C(t)$.

According the governing equations of resistance and capacitor, we describe the input-output relationship between $x(t)$ and $y(t)$ using a differential equation,

$$RC \frac{dy(t)}{dt} + y(t) = x(t) \quad (1)$$

¹<https://www.python.org>

²<https://jupyter.org>

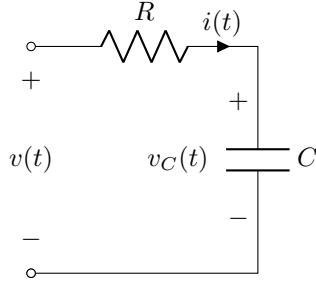


Figure 1: Series RC Circuit

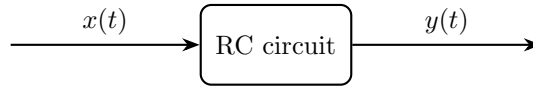


Figure 2: Basic System Block Diagram

where RC is called the time constant of RC circuit, denoted by $\tau = RC$. The system is with zero rest.

With this background, the goal of this lab is to determine the system parameter (τ) using some measurements of the system (e.g., $x(t)$ and $y(t)$), if possible. Below outlines two objectives of this lab.

- (i) Estimate the time constant τ from measured input-output signals;
- (ii) Estimate the time constant τ from measured input-output signals if they are noisy.

1 Task A Determine the time constant τ from an input-output pair

We observe an input-output pair,

- Input signal $x(t) = u(t)$;
- Output signal $y(t) = (1 - e^{-t/2})u(t)$.

We next use three approaches to determine the time constant τ .

1.1 Estimate τ using the system equation

Since the input-output pair $(x(t), y(t))$ is generated by the system (1), it is straightforward to compute the time constant τ via

$$\tau \frac{dy(t)}{dt} = x(t) - y(t).$$

From the output signal $y(t)$,

$$\frac{dy(t)}{dt} = \frac{d}{dt}(1 - e^{-t/2}) = \frac{1}{2}e^{-t/2}.$$

Therefore,

$$\tau = \frac{x(t) - y(t)}{\frac{dy(t)}{dt}} = 2.$$

In the .ipynb file, all continuous-time signals are represented as discrete-time signals and we use **numpy.gradient** to approximately compute the first-order derivative $y'(t)$. Check the function of **numpy.gradient**, and fill in the missing part in the code to compute τ .

1.2 Estimate τ using Fourier transform

To simulate the input and output signals in continuous time, we have to discretize them over a finite interval $[0, T]$. Over the interval $[0, T]$, we choose N points: $t_0 = 0, t_1, \dots, t_{N-2}, t_{N-1} = T$ and the total number of points is N . Thus, we have the input and output signals in discrete time: $x(t_k)$ and $y(t_k)$ for $k = 0, 1, \dots, N-1$. Application of discrete-time Fourier transform to $x(t_k)$ and $y(t_k)$ yields

$$X(e^{jw}) = \sum_{k=0}^{N-1} x(t_k)e^{-jk\omega}$$

$$Y(e^{jw}) = \sum_{k=0}^{N-1} y(t_k)e^{-jk\omega}.$$

The Fourier transform of the derivative of the output signal $y(t)$ is given by

$$\bar{Y}(e^{jw}) = \mathcal{F}\left(\frac{dy(t)}{dt}\right) = \sum_{k=0}^{N-1} y'(t_k)e^{-jk\omega}.$$

where $y'(t)$ is the derivative of $y(t)$. Now, the system (1) becomes, in frequency domain,

$$\tau\bar{Y}(e^{jw}) + Y(e^{jw}) = X(e^{jw}).$$

A naive approach is to fix the frequency $w = w_0$ and estimate the time constant τ via

$$\tau = \frac{|X(e^{jw_0}) - Y(e^{jw_0})|}{|\bar{Y}(e^{jw_0})|}.$$

However, if you apply this method, you will observe that the estimate is not reliable, depending the chosen frequency w_0 . Instead, we pick a set of frequencies $\{w_i\}_{i=0}^{I-1}$ to form a linear system $Ax = b$,

$$\tau\bar{Y}(e^{jw_0}) = X(e^{jw_0}) - Y(e^{jw_0})$$

$$\begin{aligned}
\tau \bar{Y}(e^{jw_1}) &= X(e^{jw_1}) - Y(e^{jw_1}) \\
&\vdots \\
\tau \bar{Y}(e^{jw_{I-1}}) &= X(e^{jw_{I-1}}) - Y(e^{jw_{I-1}})
\end{aligned}$$

equivalently, $x = \tau$,

$$A = \begin{bmatrix} \bar{Y}(e^{jw_0}) \\ \vdots \\ \bar{Y}(e^{jw_{I-1}}) \end{bmatrix}, b = \begin{bmatrix} X(e^{jw_0}) - Y(e^{jw_0}) \\ \vdots \\ X(e^{jw_{I-1}}) - Y(e^{jw_{I-1}}) \end{bmatrix}.$$

Therefore,

$$\tau = \frac{A^* A}{A^* b}.$$

In the .ipynb file, we use `numpy.fft.fft` to perform the discrete-time Fourier transform. Check the function of `numpy.fft.fft`, and fill in the missing part in the code to compute τ .

1.3 Estimate τ using GenAI tools

Come up with prompts and ask your favorite language model to determine τ given the input-output pair. Report the name of model and access time, your prompt, the response, and your summary.

2 Task B Determine the time constant τ from a noisy input-output pair

Often the case, the measurement of output is noisy. We next observe a noisy input-output pair,

- Input signal $x(t) = u(t)$;
- Output signal $y_{\text{noisy}}(t) = (1 - e^{-t/2})u(t) + n(t)u(t)$.

Here $n(t)$ is the continuous white noise with zero mean and σ^2 variance. If you are curious, $n(t)$ is called Brownian motion, which is a fundamental concept in the stochastic system analysis.

We next repeat three approaches to determine the time constant τ under a set of noise variances,

$$\sigma^2 \in \{0.0, 0.0001, 0.0002, 0.002, 0.01, 0.05\}. \quad (2)$$

2.1 Estimate τ using the system equation

We pretend that the input-output pair $(x(t), y_{\text{noisy}}(t))$ was generated by the system (1). Thus, they should satisfy

$$\tau \frac{dy_{\text{noisy}}(t)}{dt} + y_{\text{noisy}}(t) = x(t).$$

However, the derivative of the noise $n(t)$ over t is not defined, since the noise $n(t)$ is nowhere differentiable. To avoid this issue, we turn the differential equation to an integral equation,

$$\tau(y_{\text{noisy}}(t) - y_{\text{noisy}}(0)) = \int_0^t (x(t') - y_{\text{noisy}}(t')) dt'.$$

where the integral is computationally feasible. Again, we pick a set of times $\{t_i\}_{i=0}^{I-1}$ to form a linear system $Ax = b$,

$$\begin{aligned} \tau(y_{\text{noisy}}(t_0) - y_{\text{noisy}}(0)) &= \int_0^{t_0} (x(t') - y_{\text{noisy}}(t')) dt' \\ \tau(y_{\text{noisy}}(t_1) - y_{\text{noisy}}(0)) &= \int_0^{t_1} (x(t') - y_{\text{noisy}}(t')) dt' \\ &\vdots \\ \tau(y_{\text{noisy}}(t_{I-1}) - y_{\text{noisy}}(0)) &= \int_0^{t_{I-1}} (x(t') - y_{\text{noisy}}(t')) dt' \end{aligned}$$

equivalently, $x = \tau$,

$$A = \begin{bmatrix} y_{\text{noisy}}(t_0) - y_{\text{noisy}}(0) \\ \vdots \\ y_{\text{noisy}}(t_{I-1}) - y_{\text{noisy}}(0) \end{bmatrix}, b = \begin{bmatrix} \int_0^{t_0} (x(t') - y_{\text{noisy}}(t')) dt' \\ \vdots \\ \int_0^{t_{I-1}} (x(t') - y_{\text{noisy}}(t')) dt' \end{bmatrix}.$$

Therefore,

$$\tau = \frac{A^* A}{A^* b}.$$

In the .ipynb file, we provide a function **cumulative_trapezoid** to approximately compute the integral of a signal. Fill in the missing part in the code to compute τ .

2.2 Estimate τ using GenAI tools

Come up with prompts and ask your favorite language model to determine τ given a noisy input-output pair. For this task, pick a noise variance σ^2 from the given range (2). Report the name of model and access time, your prompt, the response, and your summary.

3 Report presentation

Please turn in an .ipynb file for the lab report. Name the file in the format of **ECE 315.Project.Tom** by replacing **Tom** by your first name. To earn full credits, please fill in the missing code in the .ipynb file, ensure the file is executable, and answer all questions in the discussion section.